

SUJITH NAIR

Lead Software Engineer

+1 (603) 790 3061
sujith.jay.nair@gmail.com
Hanover, NH, United States

Staff-level data platform engineer with 10+ years architecting and scaling data infrastructure for hypergrowth organizations. Led platform evolution at Nubank through 40x customer growth (3M to 140M users) across 5 countries, designing batch processing systems processing 400,000+ daily jobs. Proven track record building self-serve data platforms, driving cross-functional compliance initiatives, and contributing to critical open-source projects (Apache Spark, Apache Iceberg, Apache Datafusion-Comet). Experienced in transforming data architectures at scale, mentoring engineering teams, and establishing technical standards across distributed organizations.

PROFESSIONAL EXPERIENCE

Nubank, Lead Software Engineer, Data Platform

McLean, VA, USA | Dec 2023 - Present

Led data platform architecture, supporting hypergrowth to 140M customers across 5 countries.

- **Led the product strategy for Nubank's Data Control Plane**, a Kubernetes-operator platform that turns data infrastructure into a declarative API across > 25 data planes across 5 geos, supporting > 8 client teams
- **Architected and led migration from Mesos/Aurora to serverless Lambda-based orchestration**, eliminating the operational burden of managing Zookeeper, Mesos, and Aurora while improving system reliability to 99.9% uptime and reducing platform operational incidents by 70%
- **Led Spark-Connect integration enabling Python-native data processing**, architecting remote execution model that allowed data scientists to submit PySpark jobs without local Spark installation while maintaining security isolation and resource governance across 400K+ existing Scala workloads
- **Spearheaded comprehensive AWS EMR evaluation initiative** involving production benchmarking across 1000+ jobs, and cost-performance analysis that informed the strategic decision not to adopt EMR, saving estimated \$75K daily in potential cost increases
- **Designed and implemented fallback-to-On-Demand capability** that automatically switches from spot to on-demand instances during capacity shortages, protecting 1200+ business-critical clusters from spot market volatility and reducing on-call incidents by 60%
- **Led Data Mesh platform initiative** as a technical lead, delivering DAG Manager, flexible scheduling system, and cross-team orchestration capabilities that enabled 15+ teams to manage complex workflows across distributed repositories
- **Established technical mentorship program** across distributed teams (Berlin, São Paulo), promoting 5 engineers to senior roles and creating bi-weekly knowledge-sharing sessions that became standard practice across data platform organization

Nubank, Senior Software Engineer, Data Platform

Berlin, Germany | Jan 2019 - Dec 2022

Led data platform evolution supporting hypergrowth from 3M to 60M customers, driving cross-functional initiatives for anomaly detection, auto-remediation, and LGPD compliance

- **Architected orchestration platform evolution** that enabled 6x increase in daily pipeline executions (from 50K to 300K+) while maintaining the span execution time (< 24 hours) for each daily run through optimized resource scheduling and autoscaling strategies
- **Architected self-serve anomaly detection platform** adopted by 50+ engineering teams across

100,000+ daily datasets, reducing data quality incidents by 70%

- **Designed and implemented diagnosis and auto-remediation system** that reduced mean time to resolution for data pipeline failures from 90 mins to 15 mins, improving platform SLAs and enabling the platform to scale without proportional headcount growth
- **Led cross-functional LGPD (GDPR) compliance** initiative spanning engineering, security, legal, and product teams, architecting privacy-compliant caching layer affecting 60M+ customer records
- **Open-source leadership:** Contributed critical performance improvements to Apache Spark, Apache Iceberg, and Apache Datafusion-Comet

Logistimo, Data Engineer II

Bangalore, India | Mar 2016 - Dec 2018

Led data infrastructure transformation for public health supply chain platform serving government immunization programs across India, Zambia, Myanmar, and Indonesia as technical lead of 3-engineer team

- **Architected and led migration from RDBMS to distributed data lake** built on Apache Spark, YARN, and Cassandra, enabling the platform to process immunization supply chain data for 50M+ citizens across 4 countries
- **Collaborated with product and field teams** to define data requirements for new markets, translating complex public health needs and regulatory requirements across countries into technical specifications

WeAreHolidays, Software Engineer

Gurgaon, India | Jul 2014 - Feb 2016

Founding engineer in a tiny team building a travel marketplace from zero to launch, establishing technical foundation and development practices.

- **Co-architected and built full-stack travel marketplace platform** from ground up, delivering end-to-end booking system spanning supplier integrations, payment processing, user management, and inventory management within 8 months to first customer transaction.
- **Collaborated directly with founders** on product roadmap and technical strategy, translating business requirements into technical solutions and making pragmatic architectural trade-offs to balance speed-to-market with scalability.

EDUCATION

Masters in Technology, Systems/Electrical Engineering

Indian Institute of Technology, BHU, Varanasi

Bachelor of Engineering, Computer Engineering

University of Mumbai

SKILLS

- **Data Processing:** Apache Spark, Apache Iceberg
- **Cloud & Infrastructure:** AWS (EMR, S3, Lambda), YARN, Mesos, Airflow, Kubernetes
- **Languages:** Scala, Java, Clojure, Python
- **Leadership:** Technical Strategy & Roadmapping, Cross-Team Collaboration, Mentorship